Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 · Functional materials and green chemistry · Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2))

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2))

1 Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2))

Study focus: Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000122 · EV0000123

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module seven on functional materials, that is, polymers.

Key details

- Today we will talk about another polymer composite that is carbon fiber.
- We talk about the synthesis, properties and applications and finally we will look at the composite of carbon fiber which is known as carbon fiber reinforced polymer.
- What are carbon fibers?
- Basically, they are fibers which are made mostly of carbon atoms, and the diameter is just about 5 to 10 micrometers.
- When you compare this diameter with the human hair, as you can see in the picture here, the human hair is much thicker as compared to the carbon fiber.
- On polymerization is known as polyacrylonitrile.
- So now the next step, how we are going to use this polyacrylonitrile to get the carbon fiber.
- So here again the first step would be polymerization which I mentioned just now.
- So all these acrylonitrile molecules are going to be placed like this because it is renal cyanide.
- The polymerization will always take across take place across the double bond.
- So you can see the chain has got formed because of the opening of this double bond.
- So that is the first step which is.
- Now the next step is we need to cyclize it.
- Remember that we want a graphite like structure.
- So the next step would be cyclization that is done at a low temperature.

- Cyclization process there will be a bond which will be formed between this carbon and this nitrogen.
- Up and you get a bond getting formed between C and N, and here it will be left with double bond.
- That's exactly what you can see in this picture, and that will result in a cyclic molecules, and therefore that is called as the cyclization process.
- Polymer now has become cyclic is constituted of cyclic chains, or you can see here cyclic rings, and these are in the form of a chain.
- So now the next step would be.
- I want a double bond here.
- Because we want to go towards a graphite like structure.
- I want a double bond here.
- How do I get it?
- That is done by the oxidative treatment.
- This is the cyclization, and the third step would be oxidative treatment, which is done at a higher temperature.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 05:00–09:59

This is done at a higher temperature.

Key details

- Obviously, it is treated with oxygen and water molecule will be eliminated, and here basically we are removing hydrogen.
- Hydrogen that is unsaturation now came up.
- we are basically causing oxidation of this.
- this small this chain polymer over here.
- So now the next step is the most important step where we are going to go towards the graphite like structure.
- Of course the last step would be where you remove all other elements except carbon because we want a graphite like structure that is made of carbon.
- So graffiti is the final process and that is done at a very high temperature and in an inert atmosphere and inert atmosphere because you're treating it taking the fibre to such a high temperature, so it will it'll become inflammable, it may catch fire.
- We got some unsaturation in these positions by oxidative treatment and finally we heated it and we got the ribbon like structures joined together to give you this type of structure and finally when it is graffitized, that's called graphitization at a very high temperature.
- Then we get what we call as the carbon fiber.
- If you remember graphite is made up of it, it has a layer structure and.
- These chains between these layers actually there is a there is just Vanderwall force of attraction.
- But we do that carbon fiber is one of the strongest material.

- In graphite we have very regular pattern these lay of carbon they are just one below the other and because they're all sp2 hybridized so we have layers like this and these chains between the chains is just one wall force and the these chains can slip against each other.
- It is known as a turbostratic arrangement.
- What is that?
- These sheets of carbon fiber are half as a leaf folded or crumpled together.
- They are all folded and crumpled so as a result that slipping of chains is not possible and this is known as the turbostratic arrangement and that gives the carbon fiber a good strength.
- Say that it is also biocompatible.
- It is a fiber which is biocompatible and it has a low coefficient of thermal expansion.
- Of course, the most used property is high strength to weight ratio.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 10:00–14:59

What would be the applications?

Key details

- It's very clear such a strong material we have.
- It is used in aerospace, road and marine transport, it is used in sporting goods and medical applications in prostesis surgery.
- Ligamentary repair of those.
- Because it is biocompatible.
- Then we also have automobile goods, novel tooling, casings and bases for electronic equipment, EMI and RF shielding brushes.
- But what we have learnt now is about the advantages.
- Of course, it is very expensive due to the large ex energy expenditure.
- Which are going to use carbon fiber, especially where a lot of strength is required.
- And one more disadvantage would be volatile byproducts are formed.
- We know that when we make a composite, there has to be a fiber that is the carbon fiber, and the matrix is the epoxy resin.
- That is how it is getting the strength.
- Of course, we know light weight and high strength to weight ratio, low coefficient of thermal expansion.
- But one new property which has come here and very important is the high electrical conductivity.
- Because we know it has a graphite like structure and we know that graphite.
- is conducting in nature.
- So because of that we find that it has high electrical conductivity.
- So in the next class now we are going to talk about a new class of polymers which are called conducting.

- Polymers so far polymers were always thought of as insulators, but yes, there is a class of polymers which are which can actually conduct as good as that of the metal, and these are known as conducting polymers.

Example or application from the lecture

- Like for example, H Cn could be formed when we are making carbon fiber from polyacrylonitrile.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9975; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...
SRC00031	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...
SRC00033	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...

C Revision and retrieval practice